INCH-POUND

MIL-P-70997 (AR)
AMENDMENT 1
19 May 1994

MILITARY SPECIFICATION

PROJECTILE, 81MM, HE, M821E1/M889E1
BODY ASSEMBLY

This Amendment forms a part of Military Specification MIL-P-70997 (AR), dated 4 May 1991, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

Delete "M821E1" and substitute "M821A1" wherever it appears in the specification.

Delete "M889E1" and substitute "M889A1" wherever it appears in the specification.

PAGE 1

2.1.1, After last military specification, add:

"MIL-S-70703 - Strand Cast Steel, Carbon and Alloy, for Projectile Applications."

PAGE 2

2.1.1, Military Standards, delete: "MIL-STD-1459 - ...Components."

PAGE 3

2.2, ASTM documents list, delete: "E-381 - ...Alloys and A-322 - Hot Rolled Alloy Steel Bars."

Delete: "A-350" and substitute "E-350."

Add: "E-340 - Standard Test Method for Macroetching Metals and Alloys."

AMSC N/A FSC 1315

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PAGE 3 (cont'd)

3.1, Delete sentences 3 and 4 in their entirety and substitute:

"For HF-1 the internal soundness of the steel supplied shall comply with MIL-S-50783, except that macroetch quality shall be in accordance with Drawing 12630579. Each heat of steel shall be macroetched in accordance with ASTM-E340 and 4.4.3.1."

PAGE 4

3.5.1, At end of paragraph (after "...4.5.5.1.") add:

"Any projectile containing a discontinuity in which the base metal cannot be seen throughout the zone shall be rejected."

PAGE 7

TABLE I, add new "Examination" category:

"Ingots
Macroetch test ...(-) ...3.1 ...Visual/4.4.3.1."

Under heading "Bar Stock", delete "Macroetch Test ...4.4.3.1."

PAGE 9

4.4.2, add new paragraph:

"c. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.8). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable."

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PAGE 10

Special B, delete: "cracked" and substitute "defective."

Special C, delete in its entirety.

PAGE 13

4.4.2.3, Major 108, delete in its entirety.

Add new Major 109 as follows:

"109 ..Wall thickness, minimum (7 zones) ...III ...3.2 ...Gage."

PAGE 15

4.4.3.1, delete in its entirety and substitute:

"4.4.3.1 Macroetch test. For steel not produced by the strand cast method, macroetch tests shall be performed on the product from the top end of the top cut and bottom end of the bottom cuts from the first, middle and last marker ingots to determine compliance with 3.1. The test pieces for macroetching shall show the full cross section of the material. The faces of the test pieces shall be suitably prepared and etched in accordance with ASTM-E340.

All materials shall carry the heat number and all top and bottom cuts shall be identified. All material from marker ingots shall carry the ingot number in addition to the heat number. The top end of the top cut and the bottom end of the bottom cut of all ingots shall be identified. All markings shall be retained on the pieces until all material in the heat which remains after removal of the discard meets all cleanliness requirements contained in this specification. If all cuts from the marker ingots show compliance with the requirements of 3.1 and the following, the entire heat will be accepted, if otherwise satisfactory. The discard from the top and bottom of each ingot shall be sufficient to provide sound steel. There shall be no less than three marker ingots in each heat. These shall be the first and last useable ingots and an ingot approximately at the middle of the pouring of the heat. The mold practice used on marker ingots shall be similar to that used on all other ingots of the heat. All ingots shall have an original discard at least equivalent to the original discard on the marker ingots.

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In the event of the failure of the top cut of any of the three marker ingots, discard shall continue from the top end of that cut until acceptable metal is obtained. In additions, the top cut from each ingot in the heat shall be macroetched and shall be cut back where necessary, until acceptable metal is obtained.

In the event of the failure of the bottom cut of any of the three marker ingots, discard shall continue from the bottom end of that cut until acceptable metal is obtained. In addition, the bottom cut from each ingot in the heat shall be macroetched and shall be cut back where necessary until acceptable metal is obtained.

In the event of the failure of the entire top or bottom cut of any ingot tested, the entire heat shall be rejected unless the steel producer elects to inspect each end of each piece produced from the heat by macroetch testing.

If the material is made by the strand cast process, then sampling, test procedures and macroetch quality requirements shall be in accordance with MIL-S-70703."

PAGE 15

4.4.3.2, sentence three (3), delete: ..."ASTM-A322 referencing"...

PAGE 22

6, Beneath "NOTES" add: "(This section contains information of general or explanatory nature that may be helpful, but is not mandatory.)"

6.3, delete "AMSMC" and substitute "SMCAR."

6.4, delete "Commander ...AMSMC-QAT-I (D)" in its entirety and substitute:

"Commander, U.S. Army Armament Research, Development and Engineering Center (ARDEC), ATTN: SMCAR-QAT-I."

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6.8, delete "6.8" and substitute "6.9."

Add new paragraph:

"6.8 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation."

Delete "Ammunition, mortar."

Delete "Body."

Delete "Projectile."

Add "Mortar" beneath "Indications."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AF81)

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